Declaration of AI assistance

Linear Algebra and its Applications

Author-prepared disclosure; not an official publisher form or signature.

Manuscript. Nilpotent-free spaces of 3 by 3 matrices over the field with three elements

Human authors. Deep Bhattacharjee.

Claude through Claude Code assisted original searches, computations, Lean development and drafting. OpenAI Codex assisted the direct algebraic five-dimensional construction proof, the quadratic-form argument in characteristic three, an independent characteristic-coefficient verifier, proof auditing, scope corrections, references and typesetting.

Human oversight. The submitting author stated on 25 October 2026 that the text is a mix of human-written and AI-generated material and that all parts have been checked and understood by a human. This records that author declaration. Human authors retain responsibility for mathematical accuracy, citations and the final version; AI systems are not human coauthors.

Verification scope. The Lean development formalises the reduction and search soundness. Two compiled evaluations remain explicit parts of its trust base. This must not be described as an audit using only the three standard axioms.

Supporting record. The separate AI_PROVENANCE.md identifies the documented tasks and tools. No unprovided model version, human signature, confidential transcript or journal approval is asserted.

Policy reference. https://www.elsevier.com/about/policies-and-standards/generative-ai-policies-for-journals

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